

SAN JOSÉ STATE UNIVERSITY

Digital Media Art · CADRE Laboratory for New Media
One Washington Square, San José, CA 95192

August 30, 2026

National Science Foundation
Pathways to Enable Secure Open-Source Ecosystems (PESOSE) Program

Re: Letter of Collaboration for the FreeMoCap PESOSE Proposal

To Whom It May Concern:

I am writing regarding our ongoing use of and collaboration with the FreeMoCap project. I am a senior lecturer with the CADRE Laboratory for New Media at San José State University, where I teach digital media art with a focus on 3D tools and animation, and where both my teaching and my studio practice run on free and open-source software.

We use FreeMoCap in two connected ways. In the classroom, it gives students a way to work directly with motion capture on accessible hardware and open-source software, and to see the whole process of recording, processing, and applying motion data rather than treating it as a black box. In my own practice, FreeMoCap is the engine of a six-camera capture studio I built from secondhand webcams at BASEMENT, an artist-run space in San Francisco, where it drives the character animation for my current film work.

That use has led us to build tools around the FreeMoCap workflow. The main one is HIHO Mocap, a free and open-source Blender add-on designed to work with FreeMoCap output and support the animation and teaching workflows we use at CADRE. Developing it has given us practical experience with how FreeMoCap can be extended for artists, students, and classrooms.

Working with students has also surfaced the practical questions that come with bringing motion capture into education: what hardware makes a reliable student setup, how a system can be packaged into a reproducible classroom kit, what documentation actually helps learners at different levels of technical experience, and how complex processing workflows can be presented clearly. The capture pipeline is now installed in our classroom at SJSU, and our students this fall are its first testers.

If the proposed PESOSE project is funded, we would be glad to keep collaborating with the FreeMoCap team on these educational and creative fronts. That could include feedback from classroom use, help evaluating documentation and student-facing workflows, exploring FreeMoCap-based kits or standardized setups for teaching environments, and continuing to develop and test open tools that connect FreeMoCap with Blender and related digital-media workflows.

We look forward to continuing to share what we learn as educators, artists, and users of FreeMoCap as these tools grow.

Sincerely,

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Relationship to proposing team: External FreeMoCap user, contributor, and educational collaborator